

Analysis of Conserved and Differential Gene Expression Patterns in Indian Cancer Patients: A Pan-cancer Study

Abstract

Background:

Cancer is a multifaceted disease characterized by uncontrolled cell growth in any part of the body. The genetic mutations and variations that drive cancer development and progression **vary widely among individuals and populations**. Pan-cancer studies aim to uncover commonalities and differences across various cancer types. Specifically, they may reveal **conserved** gene expression patterns fundamental to cancer biology and identify **differential** gene expression patterns that may be unique to specific cancer types or populations. Indian Council of Medical Research (ICMR) aims to investigate various types of cancers, including breast cancer, renal cancer, colon cancer, lung cancer, and prostate cancer, which have become a growing concern in recent years. This research will facilitate early detection of each cancer type, thereby reducing the fatality rate.

Methods:

The data was taken from <https://www.kaggle.com/datasets/shibumohapatra/icmr-data/code>. The dataset consists of 801 samples from 801 individuals diagnosed with various types of cancer. Each sample contains expression values for over 20,000 genes. Statistical tools used Mean-variance summary, low-varying covariate filtering, missing data cleaning, visual summaries of differential expression (scatter plot, boxplot etc), univariate multi-class logistic regression, principal component analysis, Cluster analysis etc.

Outcome:

From the box plot it observe that the gene AGR2 is expressing high across adenocarcinomas (breast adenocarcinoma, colon adenocarcinoma, prostate adenocarcinoma and Lung adenocarcinoma) except carcinoma (kidney renal clear cell carcinoma). From the top genes driving principal component 1, Kidney renal clear cell carcinoma (KIRC) is distinguished from the rest of the cancers by a cluster of 20 genes which are consistently strongly expressed in this cancer but not as strong in the others. On the other hand, the genes AGR2, FOXA1, and SPDEF are exhibiting low to no expression in the KIRC samples but relatively higher expression levels in the other cancers – specifically PRAD. These 23 genes have a profound effect on the first principal component, which results in the separation of KIRC from the other samples in the PCA scatterplot.

Conclusion:

This study underscores such variability across cancers by assessing gene expression levels in Indian cancer patients from five distinct cancer types. We clearly observe differential patterns in specific genes and cluster of genes which make certain tissue-specific cancers different from the others (such as adenocarcinomas being different from other types of carcinomas). Using statistical techniques like PCA and clustering, we perform dimension reduction in high-dimensional expression dataset to identify key genes and gene groups of interest. We use modern visualization techniques to illustrate such differences. As a future direction, it would be interesting to explore the same comparisons in an integrative setting – combining data from other sources such as copy number alterations and protein expressions alongside gene expressions.